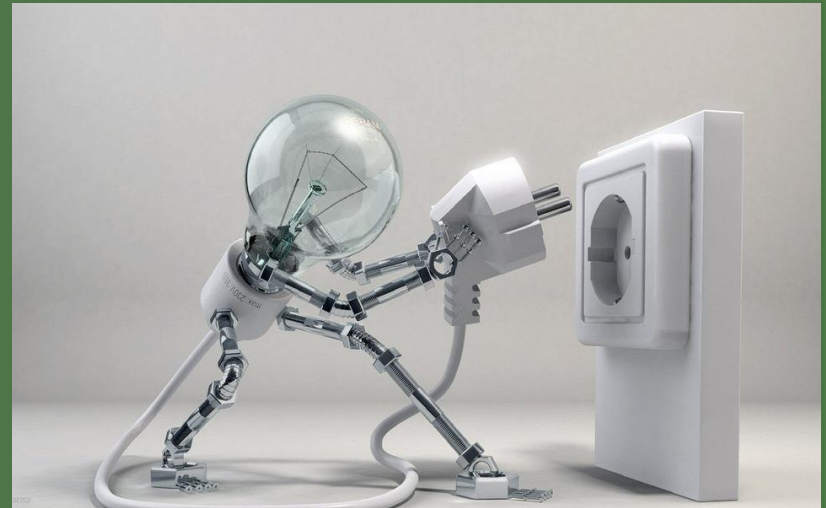


EBN111

Electricity and Electronics

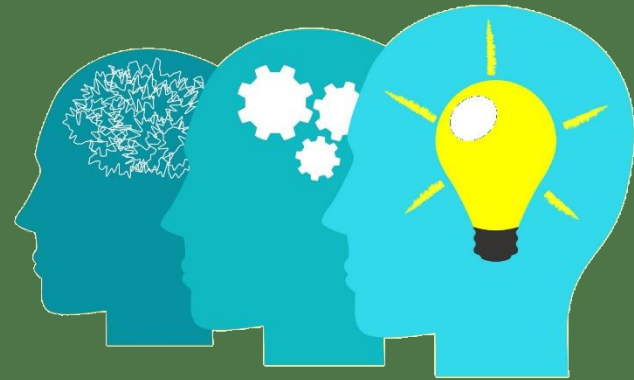
EBN111

- Why electronics and electricity?
 - Understand new concepts and theories
 - Understand implications of theory
 - Think critical about the problem
 - Apply theory to a new problem
 - Solve the problem!
 - Enjoy the challenge to solve new problems.



Aim of course

- Develop skills to solve electronic circuits
- Solve new problems,
 - in a creative manner,
 - based on theory.
- There are no recipes!
- The emphasis is on **understanding** and not on ~~memorising~~.



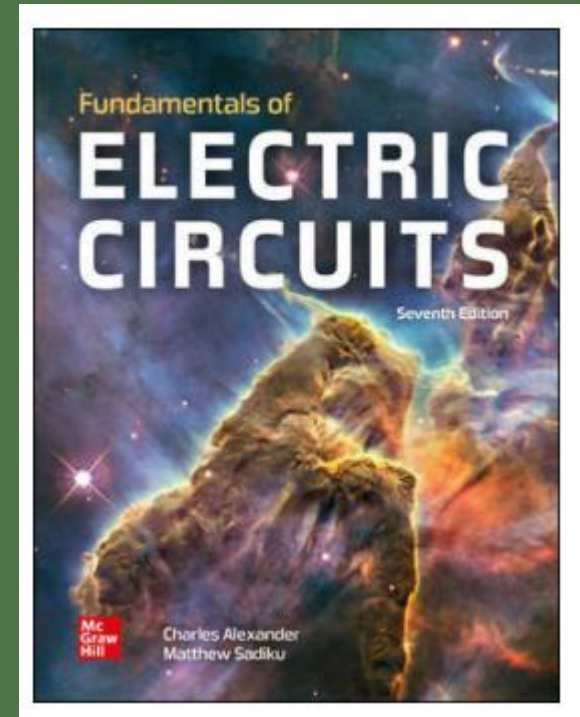
Approach

“For the sole true end of education is simply this: to teach men and women how to learn for themselves; and whatever instruction fails to do this is effort spent in vain.”

– Dorothy Sayers

Textbook

- Fundamentals of Electric Circuits
 - Seventh Edition
 - Charles K. Alexander and Matthew N. O. Sadiku
 - McGraw Hill International Edition
 - Copyright: 2021
- Famous quote:
 - *“Read the damn textbook!” - Anon*



Learning Activities

- 3 Lectures per week
 - Read work before (~5min) & after (~15min) lecture
 - Schedule time to study what you don't understand.
 - Go for consultations if you can't resolve problems.
- 1 Tutorial class per week
- Assignments for each chapter.
- ± 5 hours study time per week excluding lectures!



Lecturing team and support

	Name	Office	Contact details	Consultation
Lecturers	Prof. Xianming Ye	Eng 1:14-5	012 420 4353 xianming.ye@up.ac.za	Please make an appointment via e-mail
	Mr. Joshua Thiselton	Eng 3:7-33	Joshua.thiselton@up.ac.za	
	Mr. Leon Staphorst		leon.staphorst@gmail.com	
Assistant lecturers	Refer to module ClickUP page			
EECE undergraduate administration	Ms Bainca Johnson Ms Mari Ferreira	Eng 3:7-18 Eng 3:7-19	eece.undergraduate@up.ac.za	
Faculty Student Advisor		Your Faculty Student Advisor can advise you on goal-setting, adjustment to university life, time management, study methods, stress management and career exploration. Book an individual consultation or attend a workshop. For other support services see Paragraph 12 in the EECE General Study Guide.		

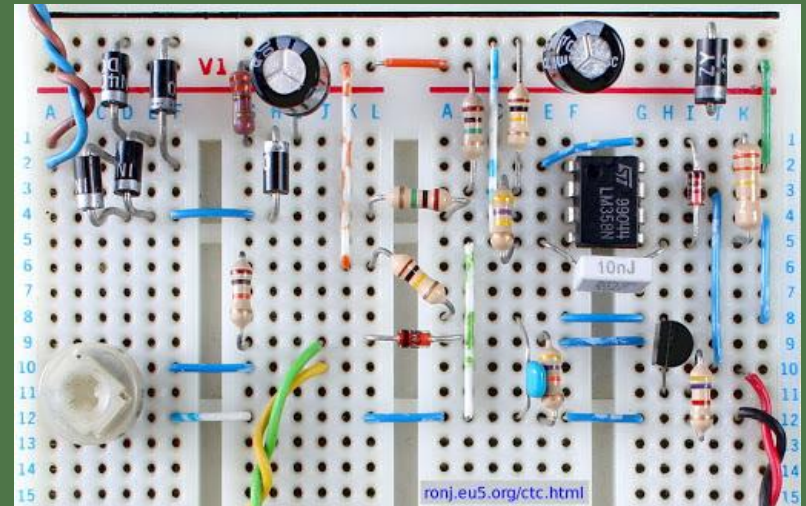
Consultations

- Consultation arrangements on ClickUP.
 - There are specific timeslots each day.
 - See ClickUP for venues for in-person consultations



Practicals

- Three practicals in the semester
 - Pre-prac lecture before every practical
 - Each practical session is 3 hours
 - You will only have 3 practical sessions in semester
- Laboratory
 - CAEC lab: Eng 2:4-39



Assessment

- Semester mark:
 - Semester tests (x2): 70%
 - Class tests (x2): 12%
 - Practicals (x3): 12%
 - AMS assignments (x9): 6%
- Voluntary third semester test
 - Best two semester tests is used to calculate semester mark.
 - No aegrotat test for third semester test!
- Exam entrance: Semester mark $> 40\%$.



Assessment

- Final mark:

Final mark (100%) = Exam mark (50%)
+ Semester mark (50%)

- Entrance to exam:

- Semester mark greater than 40%.
- All practicals must be complete.
- Average of at least 50% overall for practicals.

- To pass:

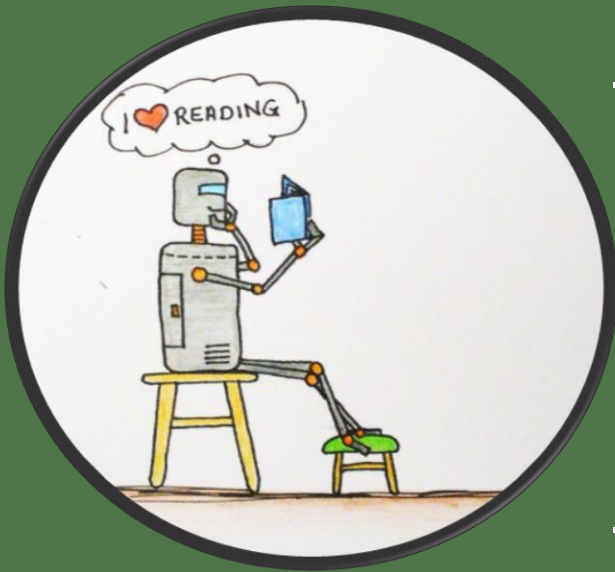
- Final mark greater or equal to 50%.



Refer to the Study Guide!

Study guide

- 8 study themes, i.e. 8 chapters
 - Study guide describes:
 - What you must do in each study theme
 - Which sections in the chapter you must do
 - Use study guide as guide on what to study!



WEEK 1: WHAT NOW?



Step 1: Access the EBN111 ClickUP module.

ebn111_s1_2024

EBN 111 S1 2024

Content Calendar Announcements Discussions Gradebook Messages Analytics Groups

Course Faculty



R (Rene) Herman

INSTRUCTOR



JD (Derik) le Roux

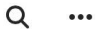
INSTRUCTOR



[Show more](#)

Details & Actions

Course Content



EBN111 key info pack

Visible to students ▼



For any EBN related queries, please email: ebn111.2024@gmail.com

Step 2: Make sure you obtain a copy of the textbook.



Study guide and lecture schedules

Visible to students ▼



All about the Prescribed Textbook

Visible to students ▼



Fundamentals of Electric Circuits

Hidden from students ▼



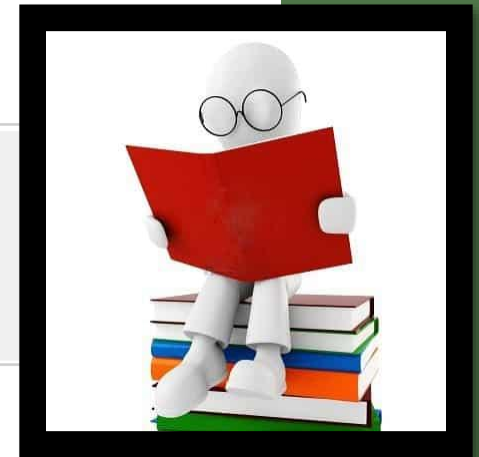
Applications for aegrotat or extraordinary assessments

Visible to students ▼



Consultation

Visible to students ▼



Step 3: Lecture Schedule

EBN111 Lecture Schedule 2024

XY – Prof. Xianming Ye; LS – Leon Staphorst; JT – Joshua Thiselton; AL – Assistant Lecturers

Venues: **Mon** - Roos Hall; **Tue** - Thuto 3-2; **Thu**: Roos hall; **Fri**: AE du Toit Auditorium

Week	Day & Date		Event	Lecture contents and study tasks
1	Mon	19/02/2024	Lecture 1	<i>Introduction and Study Guide</i> Study Ch 1.1-1.4 and do textbook Examples: 1.1, 1.2, 1.3 <i>Units, Charge, Current, Voltage</i> Do textbook Practice Problems: 1.1, 1.2, 1.3
	Tue	20/02/2024	Lecture 2	Study Ch 1.5-1.6 and do textbook Examples: 1.4, 1.5, 1.6, 1.7 <i>Power, Energy and Circuit Elements</i> Do textbook Practice Problems: 1.4, 1.5, 1.6, 1.7
	Wed	21/02/2024	Assignment 1	Do AMS Assignment #1 (Deadline: 26 February 2024 at 11:59pm).
	Thu	22/02/2024	Lecture 3	Study Ch 2.2-2.3 and do textbook Examples: 2.1, 2.2, 2.3, 2.4 <i>Ohm's Law, Nodes & Branches & Loops</i> Do textbook Practice Problems: 2.1, 2.2, 2.3, 2.4
	Fri	23/02/2024	Tut class #1	AMS tutorial Do chapter problems: 1.4, 1.6, 1.8, 1.11, 1.12, 1.16, 1.18, 1.20, 1.24, 1.28, 2.6, 2.7

Step 4: Tutorials

Tutorial days

1. Do questions at the end of the chapter.
2. The memo is released on day of tutorial.
3. View the memo and watch videos for questions you do not understand.

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Step 5: Tutorials

Where to find the tutorial problems memos?



Tutorial problems & solutions

⊘ Hidden from students ▼



Tutorial Problems & Solutions

⊘ Hidden from students ▼

Step 6: Assignments



1. Assignments are done via <https://ams.up.ac.za/>
2. Assignments will be released on AMS on Wednesday 21 Feb.

EBN111 Lecture Schedule 2024

XY – Prof. Xianming Ye; LS – Leon Staphorst; JT – Joshua Thiselton; AL – Assistant Lecturers

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Some tips for communication

Struggling with a topic

- “I don’t understand Chapter 1”
 - Too Vague
- “I am struggling with Ohm’s Law”
 - Still too vague

Email Etiquette

- Greet:
 - Dear Sir/Madam
 - Dear Mr/Mrs/Dr/Prof
- State your problem **POLITELY!!!!!!**
- Let us know who you are
Kind regards,
Name: Name Surname
Student No: 12345678

Email Etiquette

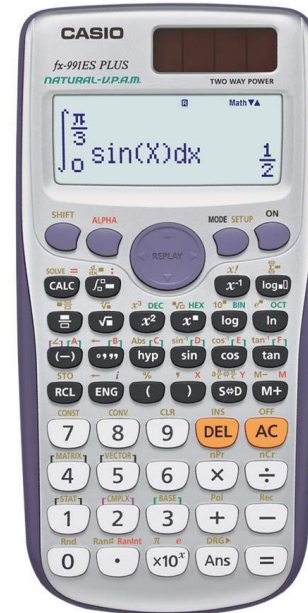
- Student who submitted a tutorial but their marks didn't appear on the list.
- Email:
 - I submitted my tutorial but my marks don't appear on the list!!!!!! Find my tutorial and rectify this mistake or I will take this to a higher authority!!!!!! My name is XYZ, please get back to me ASAP!!!!!!!!!!

Email Etiquette

- Example:
 - Dear Dr. Ye
 - The marks for tutorial 3 have been published and my marks are not on the list. I did hand in the tutorial so I am not sure why my marks have been omitted. I would appreciate your help with this problem.
 - Kind Regards,

Calculator

- Solve simultaneous equations
- Complex number
- Casio fx-991ES PLUS
- No graphing calculators!!!!



ECSA



- What is ECSA?
 - Engineering Council of South Africa
 - ECSA is in partnership with the State and the engineering profession to **promote a high level of education and training of practitioners** in the engineering profession so as **to facilitate full recognition of professionalism** in the engineering profession, both **locally and abroad**.
- What is the purpose of accreditation?
 - Engineering programmes accredited by ECSA are **recognised** as meeting the initial **academic requirements** for registration as a **Professional Engineer** in South Africa.
 - **International recognition** of these programmes is provided through the Washington Accord.
 - Signatories to the Washington Accord have undertaken to make every reasonable effort to ensure that bodies responsible for Professional Engineers practicing in its country, mutually recognise the substantial academic equivalence of engineering academic programmes accredited by other Accord signatories.

ECSA Graduate Attributes

- What are Graduate Attributes(GA's)?
 - Each engineering student is assessed according to 10 ECSA GA's at end of study period
 - GA 1: Problem Solving
 - GA 2: Application of scientific and engineering knowledge
 - GA 3: Engineering Design
 - GA 4: Investigations, experiments and data analysis
 - GA 5: Engineering methods, skills and tools, including Information Technology

ECSA Graduate Attributes

- What are Graduate Attributes(GA's)?
 - Each engineering student is assessed according to 10 ECSA GA's at end of study period
 - GA 6: Professional and technical communication
 - GA 7: Impact of Engineering activity
 - GA 8: Individual, team and multidisciplinary working
 - GA 9: Independent learning ability
 - GA 10: Engineering Professionalism

ECSA Graduate Attributes

- Graduate Attributes assessed in EBN111/122:
 - GA 1: Problem Solving
 - Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.
 - GA 2: Application of scientific and engineering knowledge
 - Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.



QUESTIONS?

